

Alejandro Román

Post-doctoral researcher

Institute of Marine Sciences of Andalusia (ICMAN-CSIC)

Date of Birth: 03/06/1996

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Education

Ph.D.	Science & Marine Technologies	ICMAN-CSIC	24/07/2024
			Qualification: Suma Cum Laude
		Awards: Extraordinary Doctoral Thesis Award from the University of Cádiz	
M.S.	Oceanography	University of Cádiz	2019
B.S.	Marine Sciences	University of Cádiz	2018

Research & Professional Experience

A. Area of Professional Expertise:

Dr. Román has extensive experience in using remote sensing technologies with drones to characterize various coastal environmental phenomena and marine ecosystems. His work includes from determining water quality parameters – such as chlorophyll-a, turbidity, and total suspended solids – through ocean color remote sensing, monitoring intertidal systems, and observing and tracking changes in polar regions. Specifically, he has worked with data obtained from various miniaturized sensors that can be mounted on drones (including RGB optical, thermal, LiDAR, gas sensors, water sampling devices, and especially multi- and hyperspectral sensors), combined with artificial intelligence techniques and programming languages (Python, R, Matlab, etc.) to develop algorithms and processing methodologies that enable the study of these systems at both local and regional scales.

B. Professional Experience:

Postdoctoral Researcher **2024/11-actually**

Institute of Marine Sciences of Andalusia (ICMAN-CSIC).

Postdoctoral researcher in the DIGIDRON project under the Momentum CSIC Programme: Develop Your Digital Talent (MMT24-ICMAN-01). The research focuses on developing emerging technologies for digitization and data processing to study environmental phenomena using drone-mounted sensors, with an emphasis on coastal systems and polar regions.

M2 contract associated with a research project **2024/08-2024/11**

Institute of Marine Sciences of Andalusia (ICMAN-CSIC).

Development of processing methodologies for the detection of sunscreen using a spectrophotometer equipped on drones, within the framework of the project “Sensors for detecting emerging contaminants for integration into observational platforms in support of sustainable tourism”.

Ph.D Candidate **2021/09-2024/07**

Institute of Marine Sciences of Andalusia (ICMAN-CSIC).

Dissertation research focused on aquatic remote sensing and optics, utilizing satellite imagery and drones to assess coastal environmental processes.

Graduate Research Assistant**2018/09-2019/07***University of Cádiz (Spain), Department of Earth Sciences.*

Development of a research project based on the use of thermal sensors onboard drones for the detection of submarine groundwater discharge (SGD) in coastal areas, with a focus on Cádiz Bay.

C. International Experience (Research Institution Stays):**University of Eastern Finland (UEF, Finland)****2025/11-2025/12***University of Eastern Finland, Joensuu, Finland.*

Stay funded under the Generation D initiative, promoted by Red.es, an organisation attached to the Ministry for Digital Transformation and the Civil Service, for the attraction and retention of talent through grants and training contracts, financed by the Recovery, Transformation and Resilience Plan through the European Union's Next Generation funds.

University of Maryland Center for Environmental Sciences (UMCES, US) **2023/08-2024/02***Horn Point Laboratory, Cambridge, MD, US.*

Stay in the framework of the Fulbright Scholarship Program.

University of Nantes (ISOMER, France)**2022/10-2022/11***University of Nantes, Nantes, France.*

Stay in the framework of the SAT4ALGAE project.

University of Malta (UM, Malta)**2022/02-2022/06***University of Malta, Msida, Malta.*

Stay in the framework of the Erasmus+KA103 Scholarship.

Kingston University (KU, United Kingdom)**2016/09-2016/12***Kingston University, London, UK.*

Stay in the framework of the Erasmus+ Studies Scholarship.

Teaching Experience

8 teaching experiences (Teaching assistant: 90 hours; Lecturer: 32 hours)

Lecturer, University of Cádiz, Spain.

2 hours of teaching in the "Ecosystems Oceanography" module (M.S. in Oceanography, University of Cádiz) – 09/12/2025.

Lecturer, University of Eastern Finland (UEF)

2 hours of teaching in the "Earth Systems Processes and Dynamics", part of the B.S. of "Geography" program (UEF) – 14/11/2025.

Lecturer, University of Cádiz, Spain.

2 hours of teaching in the "Management of the natural environment from an ecosystem-based approach" module (M.S. in Conservation, University of Cádiz) – 30/09/2025.

Lecturer, Chile Mayor University, Santiago de Chile, Chile

22 hours of teaching in the M.S. of "Remote Sensing" (Chile Mayor University) – December 2024.

Lecturer, University of Cádiz, Spain.

2 hours of teaching in the "Ecosystems Oceanography" module (M.S. in Oceanography, University of Cádiz) – 10/12/2024.

Lecturer, University of Cádiz, Spain.

2 hours of teaching in the "Management of the natural environment from an ecosystem-based approach" module (M.S. in Conservation, University of Cádiz) – 29/10/2024.

Teaching Assistant, University of Cádiz, Spain.

45 hours of teaching in the “GIS and Remote Sensing” module (2nd year at the B.S. in Marine Sciences, University of Cádiz) – Winter 2022/23.

Teaching Assistant, University of Cádiz, Spain.

45 hours of teaching in the “GIS and Remote Sensing” module (2nd year at the B.S. in Marine Sciences, University of Cádiz) – Winter 2021/22.

Mentoring

7 mentoring experiences (M.S. thesis: 6; Ph.D. research stays: 1)

Aurora Lucía Navarro Miragaya, M.S. student **2025/11-actually**

M.S. in Conservation and Management of the Natural Environment, University of Cádiz, Spain.

Research Line: Use of artificial intelligence to study the dynamics and trends of Antarctic penguin colonies.

Sergio Obrador Gómez, M.S. student **2025/11-actually**

M.S. in Conservation and Management of the Natural Environment, University of Cádiz, Spain.

Research Line: Use of drones and artificial intelligence to study the spatiotemporal variability of marine macrophyte meadows in the Bay of Cádiz.

Daniel Febles Medina, M.S. student **2025/11-actually**

M.S. in Conservation and Management of the Natural Environment, University of Cádiz, Spain.

Research Line: Study of the spatiotemporal evolution of the marsh system in the Bay of Cádiz using artificial intelligence and data collected by drone-mounted sensors.

Iván Alonso Nieto, M.S. student **2025/11-actually**

M.S. in Conservation and Management of the Natural Environment, University of Cádiz, Spain.

Research Line: Analysis of seagrass meadows in the Bay of Cádiz using drone–satellite upscaling through machine learning.

Claudia Giménez Poblador, Ph.D. student **2025/03-2025/04**

Research stay in the context of the NetOPS 2025 mobility grant calls.

Research Line: Application of SfM photogrammetry with RGB, multispectral, LiDAR, and thermal-based imagery to produce high-resolution orthomosaics in different study basins, with a focus on polar research.

María Barbudo Compagnucci, M.S. student **2025/02-actually**

M.S. in Conservation and Management of the Natural Environment, University of Cádiz, Spain.

Research Line: Use of drones and artificial intelligence for the detection and identification of macroplastics during the Cut-Off Low emergency (Valencia).

Grace Carolina Yañez Cruz, M.S. student **2025/01-2026/02**

M.S. in Oceanography, University of Cádiz, Spain.

Research Line: Integrated use of machine learning techniques and drone-based remote sensing for the study of marine organism populations in coastal areas. Qualification: 7.5.

Publications

A. Publications in Peer-Reviewed Scientific Journals:

21 publications (first author: 12; book chapters: 2; number of citations: 468 (Google Scholar); h-index:12)

1. **Román, A.**, Navarro, G., Barbero, L., Fernández-Marín, B., García-Plazaola, J.I., González-Ortegón, E., Caballero, I., Tovar-Sánchez, A. (2026) Unveiling the large coverage of red snow algae blooms in antarctic coastal snowfields. *Communications Earth & Environment*, 7, 53. DOI: 10.1038/s43247-025-03156-6

2. **Román, A.**, Tovar-Sánchez, A., Larrad, M., Rubiano-Sánchez, J., Zafra, J.M., Piñeiro, R., Castillo, A., López, F.A., Vela, A.L., Allende, A., Sánchez, G., Martínez-Alonso, A., Samper, D., García, J.C., Galindo, I., Navarro, G. (2025) UAV imagery in natural disasters: Real-time damage

assessment of flash flooding events. *Ecological Informatics*, 91, 103433. DOI: 10.1016/j.ecoinf.2025.103433

3. **Román, A.**, Tovar-Sánchez, A., Larrad, M., Alva, F., Navarro, G. (2025) LiDAR-based topographic data for the coastline of Port Foster (Deception Island, Antarctica). *Scientific Data*, 12,1363. DOI: 10.1038/s41597-025-05726-x

4. Amaya-Vías, S., Flecha, S., **Román, A.**, Haro, S., Oviedo, J.L., Navarro, G., Arroyo, G.M., Huertas, I.E. (2025) Air-water CO₂ exchange in transformed saltmarshes for different uses and under various management models. *Journal of Environmental Management*, 380, 125188. DOI: 10.1016/j.jenvman.2025.125188

5. **Román, A.**, Oiry, S., Davies, B.F.R., Rosa, P., Gernez, P., Tovar-Sánchez, A., Navarro, G., Méléder, V., Barillé, L. (2024) Mapping intertidal microphytobenthic biomass with very high-resolution remote sensing imagery in an estuarine system. *Science of the Total Environment*, 177025. DOI: 10.1016/j.scitotenv.2024.177025

6. Belyaev, O., **Román, A.**, Belliure, J., Navarro, G., Tovar-Sánchez, A. (2024) Assessing Topographic Features and Population Abundance in an Antarctic penguin colony through UAV-Based Deep-Learning Models. *International Journal of Applied Earth Observation and Geoinformation*, 134, 104124. DOI: 10.1016/j.jag.2024.104124

7. Davies, B.F.R., Oiry, S., Rosa, P., Zoffoli, M.L., Sousa, A.I., Thomas, O.R., Smale, D.A., Austen, M.C., Biermann, L., Attrill, M.J., **Román, A.**, Navarro, G., Barillé, A.L., Harin, N., Clewley, D., Martínez-Vicente, V., Gernez, P., Barillé, L. (2024) Intertidal seagrass extent from Sentinel-2 time-series show distinct trajectories in Western Europe. *Remote Sensing of Environment*, 312, 114340. DOI: 10.1016/j.rse.2024.114340

8. Davies, B.F.R., Oiry, S., Rosa, P., Zoffoli, M.L., Sousa, A.I., Thomas, O.R., Smale, D.A., Austen, M.C., Biermann, L., Attrill, M.J., **Román, A.**, Navarro, G., Barillé, A.L., Harin, N., Clewley, D., Martínez-Vicente, V., Gernez, P., Barillé, L. (2024) A sentinel watching over inter-tidal seagrass phenology across Western Europe and North Africa. *Communications Earth & Environment*, 5, 382. DOI: 10.1038/s43247-024-01543-z

9. **Román, A.**, Navarro, G., Tovar-Sánchez, A., Zarandona, P., Roque-Atienza, D., Barbero, L. (2024) ShetlandsUAVmetry: unmanned aerial vehicle-based photogrammetric dataset for Antarctic environmental research. *Scientific Data*, 11, 202. DOI: 10.1038/s41597-024-03045-1

10. **Román, A.**, Heredia, S., Windle, A.E., Tovar-Sánchez, A., Navarro, G. (2024) Enhancing Georeferencing and Mosaicking Techniques over Water Surfaces with High-Resolution Unmanned Aerial Vehicle (UAV) imagery. *Remote Sensing*, 16, 290. DOI: 10.3390/rs16020290

11. **Román, A.**, Tovar-Sánchez, A., Fernández-Marín, B., Navarro, G., Barbero, L. (2023) Characterization of an Antarctic Penguin Colony Ecosystem using High-Resolution UAV Hyperspectral Imagery. *International Journal of Applied Earth Observation and Geoinformation*, 125, 103565. DOI: 10.1016/j.jag.2023.103565

12. **Román, A.**, Prasyad, H., Oiry, S., Davies, B.F.R., Brunier, G., Barillé, L. (2023) Mapping intertidal oyster farms using unmanned aerial vehicles (UAV) high-resolution multispectral data. *Estuarine, Coastal and Shelf Science*, 291, 108432. DOI: 10.1016/j.ecss.2023.108432

13. **Román, A.**, Tovar-Sánchez, A., Gauci, A., Deidun, A., Caballero, I., Colica, E., D'Amico, S., Navarro, G. (2023) Water-Quality Monitoring with a UAV-Mounted Multispectral Camera in Coastal Waters. *Remote Sensing*, 15(1), 237. DOI: 10.3390/rs15010237

14. Bonnet-Dunbar, M., **Román, A.**, Caballero, I., Navarro, G. (2022) Remote Sensing: Satellite and RPAS (Remotely Piloted Aircraft System). In: *Marine Analytical Chemistry*, Springer. DOI: 10.1007/978-3-031-14486-8

15. Roca, M., Bonnet-Dunbar, M., **Román, A.**, Caballero, I., Zoffoli, L., Gernez, P., Navarro, G. (2022) Monitoring the Marine Invasive Alien Species *Rugulopteryx okamurae* using Unmanned Aerial Vehicles (UAVs) and Satellites. *Frontiers in Marine Science*, 9, 1004012. DOI: 10.3389/fmars.2022.1004012

16. **Román, A.**, Navarro, G., Caballero, I., Tovar-Sánchez, A. (2022) High-spatial resolution UAV multispectral data complementing satellite imagery to characterize a chinstrap penguin colony ecosystem on Deception Island (Antarctica). *GIScience & Remote Sensing*, 59(1), 1159-1176. DOI: 10.1080/15481603.2022.2101702
17. **Román, A.**, Tovar-Sánchez, A., Roque-Atienza, D., Huertas, I.E., Caballero, I., Fraile-Nuez, E., Navarro, G. (2022) Unmanned Aerial vehicles (UAVs) as a tool for Hazard assessment: The 2021 eruption of Cumbre Vieja volcano, La Palma Island (Spain). *Science of the Total Environment*, 843, 157092. DOI: 10.1016/j.scitotenv.2022.157092
18. Caballero, I., **Román, A.**, Tovar-Sánchez, A., Navarro, G. (2022) Water quality monitoring with Sentinel-2 and Landsat-8 satellites during the 2021 volcanic eruption in La Palma (Canary Islands). *Science of the Total Environment*, 822, 153433. DOI: 10.1016/j.scitotenv.2022.153433
19. Tovar-Sánchez, A., **Román, A.**, Roque-Atienza, D., Navarro, G. (2021) Applications of unmanned aerial vehicles in Antarctic environmental research. *Scientific Reports*, 11, 21717. DOI: 10.1038/s41598-021-01228-z
20. **Román, A.**, Tovar-Sánchez, A., Olivé, I., Navarro, G. (2021) Using a UAV-Mounted Multispectral Camera for the Monitoring of Marine Macrophytes. *Frontiers in Marine Science*, 8, 722698. DOI: 10.3389/fmars.2021.722698
21. Navarro, G., **Román, A.**, Roque, D., Tovar-Sánchez, A. (2021) UAV imagery in Deception Island (Antarctica): PiMetAn Antarctic campaign 2020-2021. *DIGITAL CSIC*. DOI: 10.20350/digitalCSIC/13850

B. Under Review / In Progress:

8 manuscripts (under review: 8; first author: 2)
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22. Windle, A.E., Gray, P.C., **Román, A.**, Heredia, S., Navarro, G., Silsbe, G.M. DroneWQ: A Python package for processing MicaSense multispectral drone imagery for aquatic remote sensing. *Under review: The Journal of Open Source Software*.
23. Caballero-Martínez, O., Amaya-Vías, S., Belyaev, O., **Román, A.**, Flecha, S., Huertas, I.E., Navarro, G., Tovar-Sánchez, A., Díez-Minguito, M. Spatio-temporal Variability of pCO₂ and its Air-Sea Fluxes in Deception Island (Antarctica) Driven by Tides, Wind, Waves, and Meltwater. *Under review: Global Biogeochemical Cycles*.
24. Del Valle, M.A., Navarro, G., **Román, A.**, Barbero, L., López-Ramírez, J.A., Chisvert, A., Peris-Pastor, G., Talone, M., Tovar-Sánchez, A. Optimization and application of a UAV-Based radiometric method for monitoring Sunscreen UV filters in coastal waters. *Under review: Drones*.
25. Navarro, G., García-Davalillo, J.C., Pecharroman, J.L., Tovar-Sánchez, A., Gumiel, J.C., Madrigal, J., Galindo, I., **Román, A.** Post-fire LiDAR dataset and 3D reconstruction of Las Médulas World Heritage Cultural Landscape (NW Spain). *Accepted for publication: Scientific Data*.
26. Tovar-Sánchez, A., Basallote, M., Bartual, A., Chisvert, A., Azorín, C., Rubio, E., **Román, A.**, Huertas, I.E., Navarro, G., Rodríguez-Romero, A. Tourism leaves chemical footprints in Antarctic coastal ecosystems. *Under review: Science Advances*.
27. Burchard-Levine, V., Jiménez, M., Sánchez-Virosta, A., Peón, J.J., Borra-Serrano, I., **Román, A.**, Rodríguez-Pérez, J., González-Pérez, I., Gómez-Giráldez, P., De Castro, A., Peña, J., Navarro, G., González-Piqueras, J., Díaz-Delgado, R., González-Dugo, M., Martín, M.P. Vicarious calibration of optical sensors onboard unmanned aerial vehicles: insights from inter-comparison experiments. *Under review: Computers and Electronics in Agriculture*.
28. **Román, A.**, Amaya-Vías, S., Díez-Minguito, M., Tovar-Sánchez, A., Navarro, G., Flecha, S., Huertas, I.E. Methane emissions from penguin colonies reshape the natural greenhouse gas budget. *Under review: Science Advances*.

29. **Román, A.**, Navarro, G., Tovar-Sánchez, A., Villoslada, M. Integrating Drone and Sentinel-2 Data to Improve Antarctic Vegetation Mapping: A Comparison of Upscaling and Super-Resolution Approaches. *Under review: Engineering Applications of Artificial Intelligence*.

C. Conference Publications:

11 proceeding publications derived from conference abstracts

1. **Román, A.**, Windle, A.E., Tovar-Sánchez, A., Navarro, G., Silsbe, G. (2025) A New AI Approach to Retrieving Ocean Color Products from Unoccupied Aircraft Systems (UAS). *In: International Ocean Colour Science Meeting 2025 Book of Abstracts*.

2. **Román, A.**, Windle, A.E., Tovar-Sánchez, A., Navarro, G., Silsbe, G. (2024) Uso de dron equipado con sensor multiespectral para la evaluación de la calidad de agua en la Bahía de Chesapeake (Estados Unidos). *In: Conference Book of the XX Congreso de la Asociación Española de Teledetección (AET)*.

3. **Román, A.**, Oiry, S., Davies, B.F.R., Rosa, P., Tovar-Sánchez, A., Navarro, G., Méléder, V., Barillé, L. (2024) Exploración de la relación cuantitativa entre la pigmentación del Microfitobentos (MPB) y las medidas derivadas del empleo de un dron equipado con sensor multiespectral. *In: Conference Book of the XX Congreso de la Asociación Española de Teledetección (AET)*.

4. Castañeda, C., López-Flores, R., **Román, A.**, Mota, D., Abdelrazek, A., Navarro, G. (2024) Drones para el muestreo de agua y el registro de datos multiespectrales en humedales protegidos de Aragón localizados en cuencas agrícolas. *In: Conference Book of the XX Congreso de la Asociación Española de Teledetección (AET)*.

5. Del Valle García, M.A., Navarro, G., **Román, A.**, Barbero, L., López, J.A., Chisvert, A., Peris, G., Tovar-Sánchez, A. (2024) Optimización y aplicación de análisis espectrofotométrico para la monitorización de filtros UV en aguas costeras. *In: Conference Book of the XX Congreso de la Asociación Española de Teledetección (AET)*.

6. **Román, A.**, Tovar-Sánchez, A., Fernández-Marín, B., Navarro, G., Barbero, L. (2024) High-Resolution UAV Hyperspectral Imagery for Antarctic Research. *In: European Geosciences Union (EGU) General Assembly (2024), Vienna (Austria), EGU24-154*.

7. **Román, A.**, Tovar-Sánchez, A., Gauci, A., Deidun, A., Caballero, I., Colica, E., D'Amico, S., Heredia, S., Navarro, G. (2023) High-resolution UAV multispectral imagery for water-quality monitoring in coastal regions. *In: European Geosciences Union (EGU) General Assembly 2023, Vienna (Austria), EGU23-8599*.

8. **Román, A.**, Tovar-Sánchez, A., Roque-Atienza, D., Huertas, I.E., Caballero, I., Fraile-Nuez, E., Navarro, G. (2022) Use of Unmanned Aerial Vehicles (UAVs) for emergency assessment during volcanic activities: La Palma Island volcanic eruption (2021). *In: Conference Book of the VIII International Symposium on Marine Sciences 2022 (ISMS)*.

9. **Román, A.**, Tovar-Sánchez, A., Caballero, I., Olivé, I., van Bergeijk, S.A., Vilas, C., Navarro, G. (2021) Uso de UAV con sensor multiespectral para la monitorización de macrófitos marinos y microfitobentos (MPB) en zonas intermareales. *In: Conference Book of the XIX Congreso de la Asociación Española de Teledetección (AET)*. ISBN: 978-84-9769-383-7

10. **Román, A.**, Tovar-Sánchez, A., Olivé, I., Navarro, G. (2021) Monitorización de Macrófitos Marinos en la Bahía de Cádiz mediante dron con sensor multiespectral. *In: Conference Book of the III JIS del Mar, Fundación CEI-Mar*. ISBN: 978-84-09-36126-7

11. **Román, A.** (2021) Methodologies for the detection and study of submarine groundwater discharges (SGD) in the Gulf of Cádiz. *In: Conference Book of the II JIS del Mar, Fundación CEI-Mar*. ISBN: 978-84-09-13664-3

D. White Papers:

Participation in the elaboration of 1 white paper

1. Xie, Z., Lamers, M., Krüger, O., Ademollo, N., Patrolecco, L., Di Cicco, A., Jahnke, A., Kallenborn, R., Manno, C., King, C.K., Van Opzeeland, I., Thomisch, K., Sciacca, V., Filiciotto, F., Filún, D., Fabris, R., Cajiao, D., Lee, J.R., Hughes, K.A., Lynch, H., Gross, S., Havermans, C., Grotti, M., Di Liberto, L., Corsolini, S., Calace, N., Kaiser, T., Nash, S.B., Meyer, B., Bahlburg, D., Arata, J.A., Friedlander, A., Johannessen, E.D., Kraft, B.A., Krüger, L., Lowther, A., Santos, M., Soutullo, A., Warwick-Evans, V., Cusick, A., Cortes, S., Saeedi, H., Dommergue, A., Angot, H., Heimbürger-Boavida, L., Muthuswamy Ponniah, J., Figuerola, B., Aves, A., **Román, A.**, Tovar-Sánchez, A., Ebinghaus, R., Küster, A. (2026) Theme 5: Anthropogenic signatures in Antarctica: the race against pollution and other pressures. *Submitted to: Antarctica InSync*.

Presentations

A. Conferences:

Participation in 17 national/international conferences with 31 communications (oral comm.: 11, poster comm.: 5, co-authoring: 14)

Oral Communications (Presenter)

1. Del Valle, M.A., Navarro, G., **Román, A.**, Barbero, L., López-Ramírez, J.A., Chisvert, A., Peris-Pastor, G., Talone, M., Tovar-Sánchez, A. (2026) Optimization and application of a radiometric method for monitoring UV filters in coastal waters. Oral presentation at Ocean Sciences Meeting, Glasgow (Scotland), 22-27/02/2026.
2. **Román, A.**, Windle, A.E., Tovar-Sánchez, A., Navarro, G., Silsbe, G. (2024) Uso de dron equipado con sensor multiespectral para la evaluación de la calidad de agua en la Bahía de Chesapeake (Estados Unidos). *Oral presentation at XX Spanish Remote Sensing Association Conference (AET), Cádiz (Spain), 04-07/06/2024*.
3. **Román, A.**, Tovar-Sánchez, A., Fernández-Marín, B., Navarro, G., Barbero, L. (2024) High-Resolution UAV Hyperspectral Imagery for Antarctic Research. *Oral presentation at European Geosciences Union (EGU) General Assembly, Vienna, 14-19/04/2024*.
4. **Román, A.**, Heredia, S., Windle, A.E., Tovar-Sánchez, A., Silsbe, G., Navarro, G. (2024) Novel Direct Georeferencing-Based Method for Accurate Water Surface Mosaicking with High-Resolution UAV Imagery. *Oral presentation at Ocean Sciences Meeting, New Orleans (US), 02/22/2024*.
5. **Román, A.**, Tovar-Sánchez, A., Roque-Atienza, D., Huertas, I.E., Caballero, I., Fraile-Nuez, E., Navarro, G. (2023) Use of Remote Sensing Techniques for Volcanic Eruption Monitoring: The 2021 Eruption on La Palma Island, Spain. *Oral presentation at Virtual II Symposium Vulcana, Las Palmas de Gran Canaria, 01/12/2023*.
6. **Román, A.**, Tovar-Sánchez, A., Gauci, A., Deidun, A., Caballero, I., Colica, E., D'Amico, S., Heredia, S., Navarro, G. (2023) High-resolution UAV multispectral imagery for water-quality monitoring in coastal regions. *Oral presentation at European Geosciences Union (EGU) General Assembly, Vienna, 24-28/04/2023*.
7. **Román, A.**, Tovar-Sánchez, A., Roque-Atienza, D., Huertas, I.E., Caballero, I., Fraile-Nuez, E., Navarro, G. (2022) Unmanned aerial vehicles (UAVs) as a tool for Hazard assessment: The 2021 eruption of Cumbre Vieja volcano, La Palma Island (Spain). *Oral presentation at Virtual I Symposium Vulcana, Las Palmas de Gran Canaria, 21/11/2022*.
8. **Román, A.**, Tovar-Sánchez, A., Caballero, I., Olivé, I., Van Bergeijk, S.A., Vilas, C., Navarro, G. (2022) Uso de UAV con sensor multiespectral para la monitorización de macrofitos marinos y microfitobentos (MPB) en zonas intermareales. *Oral presentation at XIX Spanish Remote Sensing Association Conference (AET), Pamplona (Spain), 23-27/07/2022*.

9. **Román, A.**, Tovar-Sánchez, A., Roque-Atienza, D., Huertas, I.E., Caballero, I., Fraile-Nuez, E., Navarro, G. (2022) Use of Unmanned Aerial Vehicles (UAVs) for Emergency Assessment during Volcanic Activities: La Palma Island Volcanic Eruption (2021). *Oral presentation at VIII International Symposium on Marine Sciences (ISMS), 04-08/07/2022.*

10. **Román, A.** (2022) Use of Unmanned Aerial Vehicles (UAV) for the study of coastal environmental processes in Antarctica (PiMetAn project). *Oral presentation at Virtual PTI-POLARCSIC Early Careers Scientists Journey, 13/05/2022.*

11. **Román, A.** (2019) Methodologies for the detection and study of submarine groundwater discharges (SGD) in the Gulf of Cádiz. *Oral presentation at II Young Researchers of the Sea Conference, 01-04/10/2019.*

Poster (Presenter)

12. **Román, A.**, Windle, A.E., Tovar-Sánchez, A., Navarro, G., Silsbe, G. (2025) A New AI Approach to Retrieving Ocean Color Products from Unoccupied Aircraft Systems (UAS). *Poster presentation at the International Ocean Colour Science Meeting 2025, Darmstadt (Germany), 01-04/12/2025.*

13. **Román, A.**, Navarro, G., Barbero, L., Fernández-Marín, B., García-Plazaola, I., González-Ortegón, E., Caballero, I., Tovar-Sánchez, A. (2025) High-Resolution UAV-Based Hyperspectral Imagery Reveals the Large Spread of Albedo-Reducing Snow Algae Blooms in Maritime Antarctica. *Poster presentation at ESA Living Planet Symposium (LPS), Vienna (Austria), 23-27/06/2025.*

14. **Román, A.**, Oiry, S., Davies, B.F.R., Rosa, P., Tovar-Sánchez, A., Navarro, G., Méléder, V., Barillé, L. (2024) Exploración de la relación cuantitativa entre la pigmentación del Microfitobentos (MPB) y las medidas derivadas del empleo de un dron equipado con sensor multiespectral. *Poster presentation at XX Spanish Remote Sensing Association Conference (AET), Cádiz (Spain), 04-07/06/2024.*

15. **Román, A.**, Navarro, G., Caballero, I., Tovar-Sánchez, A. (2022) Chinstrap penguin colonies monitoring at Deception Island (Antarctica) with high-resolution UAV imagery and moderate-resolution satellite imagery. *Poster presentation at ESA Living Planet Symposium (LPS), 23-27/05/2022.*

16. Dunbar, M.B., **Román, A.**, Caballero, I., Zoffoli, M.L., Gernez, P., Navarro, G. (2022) Monitoring the invasive alien species *Rugulopteryx okamurae* using unmanned aerial vehicles and satellites: a zoomed-out approach to biodiversity monitoring. *Poster presentation at ESA Living Planet Symposium (LPS), 23-27/05/2022.*

Co-Authoring

17. Febles, D., **Román, A.**, Xanxo-Prilló, R., Navarro, G., Haro, S. (2026) Mapping intertidal microphytobenthos using machine learning on multispectral drone imagery. *Poster presentation at the EuroMarine Open Science Day 2026, Cádiz (Spain), 10-12/03/2026.*

18. Navarro-Miragaya, L., **Román, A.**, Tovar-Sánchez, A., Navarro, G. (2026) Use of machine learning models and unmanned aerial vehicle data for the spatio-temporal analysis of chinstrap penguin colonies on Deception Island (Antarctica). *Poster presentation at the EuroMarine Open Science Day 2026, Cádiz (Spain), 10-12/03/2026.*

19. Xanxo-Prilló, R., **Román, A.**, Roca, M., Tovar-Sánchez, A., Navarro, G. (2026) Applications of Unmanned Aerial Vehicles for Coastal Marine Monitoring. *Poster presentation at the EuroMarine Open Science Day 2026, Cádiz (Spain), 10-12/03/2026.*

20. Obrador, S., Xanxo-Prilló, R., Navarro, G., **Román, A.** (2026) Spatio-temporal characterization of seagrass meadows using UAV multispectral data and machine learning models. *Poster presentation at the EuroMarine Open Science Day 2026, Cádiz (Spain), 10-12/03/2026.*

21. Navarro, G., **Román, A.**, Tovar-Sánchez, A. (2026) Use and Applications of UAV in Polar Research. *Poster presentation at the EuroMarine Open Science Day 2026, Cádiz (Spain), 10-12/03/2026.*
22. Guzmán-Morales, A.L., **Román, A.**, Suklje, L.M. (2025) Evaluation of Satellite Ocean Color Products across the North Atlantic: Insights from the One Ocean Expedition 2025. *Poster presentation at the International Ocean Colour Science Meeting 2025, Darmstadt (Germany), 01-04/12/2025.*
23. Belyaev, O., **Román, A.**, Belliure, J., Navarro, G., Barbero, L., Tovar-Sánchez, A. (2025) Evaluating topographic characteristics and population density in an Antarctic penguin colony using UAV-driven deep learning models. *Poster presentation at ESA Living Planet Symposium (LPS), Vienna (Austria), 23-27/06/2025.*
24. Castañeda, C., López-Flores, R., **Román, A.**, Mota, D., Abdelrazek, A., Navarro, G. (2024) Drones para el muestreo de agua y el registro de datos multiespectrales en humedales protegidos de Aragón localizados en cuencas agrícolas. *Oral presentation at XX Spanish Remote Sensing Association Conference (AET), Cádiz (Spain), 04-07/06/2024.*
25. Del Valle García, M.A., Navarro, G., **Román, A.**, Barbero, L., López, J.A., Chisvert, A., Peris, G., Tovar-Sánchez, A. (2024) Optimización y aplicación de análisis espectrofotométrico para la monitorización de filtros UV en aguas costeras. *Poster presentation at XX Spanish Remote Sensing Association Conference (AET), Cádiz (Spain), 04-07/06/2024.*
26. Peterson, S., Spera, A., **Román, A.**, Loughheed, V., Tweedie, C.E. (2024) Improving the understanding of spatiotemporal controls on nearshore water quality in an Arctic Lagoon. *Oral presentation at Ocean Sciences Meeting, New Orleans (US), 02/19/2024.*
27. Davies, B.F.R., Oiry, S., Rosa, P., Zoffoli, M.L., Sousa, A.I., Thomas, O.R., Smale, D.A., Austen, M.C., Biermann, L., Attrill, M.J., **Román, A.**, Navarro, G., Barillé, A.L., Harin, N., Clewley, D., Martínez-Vicente, V., Gernez, P., Barillé, L. (2023) A Sentinel Watching Over Intertidal Seagrass Phenology. *Poster presentation at EC-ESA Joint Earth System Science Initiative, Frascati, Rome (Italy), 23/11/2023.*
28. Ferrera, I., Salles, S., Cabello, A.M., Martín, J.P., González-Vega, A., **Román, A.**, Huertas, I.E., Tovar-Sánchez, A., Arrieta, J.M., Fraile-Nuez, E. (2023) Impacts of the Volcanic Eruption in La Palma (Canary Islands) on the Nearby Marine Bacterioplankton Communities. *Oral presentation at ASLO Aquatic Sciences Meeting, Palma de Mallorca (Spain), 04-09/06/2023.*
29. Roca, M., Dunbar, M.B., **Román, A.**, Caballero, I., Zoffoli, L., Gernez, P., Navarro, G. (2022) Monitoring the marine invasive alien macrophyte *Rugulopteryx okamurae* using remote sensing in Tarifa (Spain). *Poster presentation at XIX Spanish Remote Sensing Association Conference (AET), Pamplona (Spain), 23-27/07/2022.*
30. Huertas, I.E., Fraile-Nuez, E., Flecha, S., Tovar-Sánchez, A., Navarro, G., Amaya-Vías, S., Rodríguez, S., **Román, A.**, Roque-Atienza, D., Caballero, I. (2022) Assessing the impact of the Cumbre Vieja volcanic eruption on the coastal fringe of La Palma Island: from the space to the sea bottom. *Oral presentation at Goldschmidt Conference, Honolulu (Hawaii), 10-15/07/2022.*

B. Invited talks:

10 invited talks (seminars: 7 hours 25 minutes; joint sessions: 3 hours 30 minutes)

1. **Román, A.** (2026) Use of drone imagery in the monitoring of Antarctic coastal ecosystems, EGU 1st Campfire Series, 26/03/2026 (45 minutes seminar).
2. **Román, A.** (2026) Use of UAV-mounted sensors for the study of penguin colonies in Antarctica, E Group Seminar Series, 24/03/2026 (1 hour seminar).
3. **Román, A.** (2026) Artificial Intelligence Applied to Drones: Upscaling vs Super-Resolution in the Study of the Bay of Cádiz, REWRITE remote sensing upscaling workshop, 13/03/2026 (15 minutes seminar).

4. **Román, A.** (2025) UAV Imagery in Natural Disasters: Real-Time Damage Assessment of Flash Flooding Events, Spanish National School of Civil Protection, 19/11/2025 (45 minutes seminar).
5. **Román, A.** (2025) My journey through the University of Cádiz, Inaugural lecture of the Degree in Marine Sciences at the University of Cádiz (UCA, Spain), 29/09/2025 (1 hour seminar).
6. **Román, A.** (2025) Use of drone-mounted sensors to study penguin colonies in Antarctica, POLARCSIC-HUB Icebreaker Meeting, 22/09/2025 (15 minutes seminar).
7. Donlon, C., Collard, F., Andrés, L., Sukjle, L., **Román, A.** (2025) ESA Advanced Ocean Training Course: Harnessing satellite data to unlock insights into our blue planet, ESA Living Planet Symposium (Vienna, Austria), 25/06/2025 (45 minutes joint table for discussion in an Agora session).
8. **Román, A.** (2025) Novel Direct Georeferencing-Based Method for Accurate Water Surface Mosaicking with UAS multispectral Imagery, FRM4drones-AQUATIC Workshop on Drone Data Acquisition and Processing Protocols (online), 11/06/2025 (20 minutes seminar and 45 minutes joint table for discussion).
9. **Román, A.** (2024) Research in the marine environment with drones and satellites (Rota, Spain), 28/11/2024 (3 hours seminar).
10. **Román, A.** (2024) From Oceans to Coasts: Advancements and Applications of UAVs and Satellite Observations: Part II – UAVs. MASS24: Mallorca Science School 2024 (Palma de Mallorca, Spain), 21/10/2024 (45 minutes seminar).
11. **Román, A.** (2024) Pre-departure orientation talk for Fulbright scholars. Oral presentation in the Pre-Departure Orientation session for Fulbright scholars for the 2024/25 academic year (Madrid, Spain), 27/06/2024 (30 minutes seminar and two hours joint table at the cultural session).

C. Seminars / Webinars:

Participation in 8 seminars/webinars with oral communications (on-site: 7; online: 1)

Román, A. (2025) Development of remote sensing tools and use of sensors onboard UAVs for the study of coastal environmental processes. *Oral presentation at UEF Department of Geographical and Historical Studies, 26/11/2025.*

Román, A. (2025) Drones in polar research: II campaign of the DICHOSO project. *Oral presentation at the ICMAN-CSIC monthly seminars, 13/03/2025.*

Román, A. (2025) Use of drones in polar research. *Oral presentation at the Spanish Antarctic Base Gabriel de Castilla Scientific Seminars, 05/02/2025.*

Román, A. (2023) From Poles to Propellers: Drones in Antarctic Research. *Oral presentation at Horn Point Laboratory (UMCES) Student Seminars, 03/11/2023.*

Román, A. (2022) Game of Drones: coastal environmental processes from the air. *Oral presentation at Earth Observation Group Seminars at ICMAN-CSIC, 08/04/2022.*

Román, A., Sparaventi, E. (2021) Pimetrivial, el juego de preguntas sobre la Antártida. *Oral presentation at Manuel de Falla High School (Puerto Real, Cádiz), 12/11/2021.*

Román, A., Tovar-Sánchez, A., Olivé, I., Navarro, G. (2021) Monitorización de Macrófitos Marinos en la Bahía de Cádiz mediante Dron con Sensor Multiespectral. *Oral presentation at III Young Researchers of the Sea Conference, 06-09/09/2021.*

Román, A. (2020) Desarrollo de herramientas de teledetección y drones para el estudio de procesos costeros. *Oral presentation at M.S. in Oceanography during a virtual session, 09/12/2020.*

D. Scientific Outreach Events:

Participation in more than 10 on-site outreach events (national: 8; international: 3)

European Researchers' Night

September 2025

Participation as a researcher at the ICMAN-CSIC Drone stand.

REWRITE project Annual Meeting

October 2024

Participation in the Horizon Europe REWRITE project Annual Meeting in Ghent (Belgium), including a short presentation in the Mixer session.

Horn Point Laboratory Open House

October 2023

Participation in the Open House at Horn Point Laboratory (HPL) on its 50th anniversary. Scientific outreach activities showcasing the work conducted at HPL.

XVII International Science on the Street Conference, DIVERCIENCIA

May 2023

Participation in the Diverciencia 2023 event at the booth of the Institute of Marine Sciences of Andalusia, presenting and showcasing drones from the SeaDron platform.

Blue Cloud Hackathon

February 2022

Participation in the Blue Cloud Hackathon 2022, hosted online, as part of the "Ocean Warning - Heat Busters" team.

Ocean Hackathon

October 2020

Participation in the Ocean Hackathon 2020, held at the Zona Franca de Cádiz, as part of the project "Detection of plastics using drones".

European Researchers' Night

September 2018

Participation as a researcher at the University of Cádiz Drone Service stand.

IV Science and Knowledge Outreach Plan

September 2017

Participation as a volunteer in three scientific outreach events: "Researchers' Night", "Science is Eaten", and "Cádiz, 300 Years of the Sea".

Leadership

A. Successful Competitive Calls:

1. **POLARIN 2nd Transnational Access Call (Ref: 10113094)** "AMIGA: Antarctic Methane and Ice-Marine-Guano Assessment", 2026-2028 (102,000€). PI: Alejandro Román.
2. **JAE INTRO ICU POLARCSIC (POLAR-01)** "ABANTI: Análisis espacio-temporal de blooms de algas en la nieve mediante teledetección con drones en ambientes costeros antárticos", 2026 (6,000€). PIs: Alejandro Román and Beatriz Fernández-Marín.

B. Participation in Research Projects:

1. **Project: RoGEO** "GEOspatial tools for monitoring the strandings of the invasive alga *Rugulopteryx okamurae* along the Andalusian coast: mitigating its impact on fisheries and global warming". Pleamar Program 2025. Biodiversity Foundation. Spanish Ministry for the Ecological Transition and the Demographic Challenge and FEMPA. ICMAN-CSIC, 2026-2028 (184,670€). PI: Sara Haro. *Work team*
2. **Project: POLAROMICS** "Biogeography and Adaptation Mechanisms in Prokaryotes Across Polar Biomes: Ecological Connectivity and Diversification Processes in Polar Microbiomes" (Ref: PID2023146-919NB-C21). Institut de Ciències del Mar (ICM-CSIC), 2024-2028 (313,750€). PI: Silvia González Acinas. *Work team*
3. **Project: REWRITE** "REWilding and Restoration of InterTidal sediment Ecosystems for carbon sequestration, climate adaptation and biodiversity support" (Ref: 101081357). University of Nantes, 2023-2028 (7,932,116€). PI: Vona Méléder. *Work team*

4. **Project: DICHOSO** “Contribution of water masses of Deception Island to biogeochemical inventories of the Southern Ocean: current budgets and future trends” (Ref: PID2021-125783OB-100). CSIC, 2021-2024 (338,800€). PI: Antonio Tovar-Sánchez and Emma Huertas. *Work team*
5. **Project: TurisDron** “Drone technology in support of sustainable tourism” (Ref: ProyExcel_00052). CSIC, 2022-2025 (157,757€). PI: Antonio Tovar-Sánchez. *Work team*
6. **Project: Sat4Algae** “Development of satellite tools for the detection of algal blooms: Macroalgae and Microalgae” (Ref: PY20_00244). CSIC, 2021-2022 (85,290€). PI: Gabriel Navarro. *Work team*
7. **Project: Sen2Coast** “Improved assessment of algorithms for bathymetry and water quality parameters with Sentinel-2A/B high-resolution satellites for advancement of coastal applications” (Ref: RTI2018-098784-J-100). CSIC, 2019-2022 (165,770€). PI: Isabel Caballero. *Work team*
8. **Project: PiMetAn** “The role of penguins in the biogeochemical cycles of trace metals in the Southern Ocean” (Ref: RTI2018-098048-B-100). CSIC, 2019-2021 (272,250€). PI: Antonio Tovar-Sánchez. *Work team*

C. Participation in Committees:

1. Coordinator of **Objective 8 of the MAX-CSIC project (DEEPMAX-2024-2)**: “Scientific Leadership and Cross-Disciplinary Technology”.
2. Core Team member of the **Antarctica InSync Working Group "Coastal Connection and Hydrographic Drivers of Antarctic Biogeochemistry and Carbon Cycling"** since December 2025.
3. Member of the **POLARCSIC Hub Coordinating Committee** since March 2025.
4. Member of the **Organizing Committee and volunteer at the 2024 Spanish Remote Sensing Association (AET) Conference** hosted in Cádiz, Spain.

Awards & Honors

A. Awards:

Extraordinary Doctoral Thesis Award from the University of Cádiz	October 2025
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B. Obtained grants and scholarships:

Fulbright Scholarship for Pre-Doctorate Studies (Maryland, US)	2023/08-2024/02
Fulbright Scholarship Pre-Academic Program (Ohio, US)	2023/07-2023/08
FPU Scholarship by the Ministry of Science and Innovation (Spain)	2021/11-2024/11
Collaboration Scholarship in University Departments by the Ministry of Education, Culture and Sports (Spain)	2018/11-2019/07

C. Conferences funding:

EC Grants European Space Agency LPS (600€) <i>ESA Living Planet Symposium (Vienna, Austria)</i>	June 2025
UCA's Own Plan 2023-2024 for Research and Transfer Support (1,000€) <i>EGU 2024 General Assembly (Vienna, Austria)</i>	April 2024
UCA's Own Plan 2023-2024 for Research and Transfer Support (1,500€) <i>Ocean Sciences Meeting 2024 (New Orleans, US)</i>	February 2024
UCA's Own Plan 2022-2023 for Research and Transfer Support (1,000€) <i>EGU 2023 General Assembly (Vienna, Austria)</i>	March 2023
UCA's Own Plan 2021-2022 for Research and Transfer Support (350€)	July 2022

ISMS 2022 (Las Palmas de Gran Canaria, Spain)

EC Grants European Space Agency LPS (700€)
ESA Living Planet Symposium (Bonn, Germany)

May 2022

D. Successful international applications:

ELLIS Summer School 2025

2025/09-2025/09

University of Jena (Germany) – Selected for the ELLIS Summer School on AI for Earth & Climate Sciences, that will be held in September 2025 in Jena (Germany).

PACE DATA HACKWEEK 2025

2025/08-2025/08

NASA – Selected for the training for *the retrieval, processing and application of PACE hyperspectral data*, that will be held in August 2025 in Baltimore (Maryland, US).

Copernicus Training In Above-Water Radiometry (FRM4SOC-2025)

2025/07-2025/07

EU Copernicus Programme – Selected for the training for *in-situ* Ocean Colour Above-Water Radiometry towards Satellite Validation, that will be held in July 2025 in Venice and nearby Acqua Alta Oceanographic Tower (AAOT) in the Adriatic Sea.

ESA Advanced Ocean Synergy Training Course 2025 (OTC25)

2025/04-2025/06

European Space Agency (ESA) – Selected for the scientific voyage Tromsø – Nice aboard the tall ship Statsraad Lehmkuhl, and a mandatory on-line course. Including a shore-based course (58 hours: 2 ECTS) and an ocean practical course (172 hours: 6 ECTS).

Fulbright Scholarship Pre-Academic Program.

2023/07-2023/08

Kent-State University (US) – Selected for the Fulbright Scholarship Pre-academic Program at Gerald H. Read Center for International and Intercultural Education (Kent, OH, US). This program orients Fulbright graduate students to US higher education and provides academic skills and communication seminars. Students also focus on leadership development in their Fulbright seminar.

GEOTRACES Summer School

2022/07-2022/07

Helmholtz Centre For Polar and Marine Research (AWI), Bremerhaven, Germany – Selected for this one-week course, which includes a combination of lectures and hands-on training in marine geochemistry of trace elements and their isotopes, with a full day of shipboard sampling of marine trace elements onboard RV Heincke.

Training / Workshops / Courses

Remote Sensing

Advanced Training Course NASA PACE

October 2024

NASA ARSET, Online

Certificate of completion for a 6-hour advanced training course entitled “Introduction to Plankton, Aerosol, Cloud, Ocean Ecosystem (PACE) Hyperspectral Observations for Water Quality Monitoring.

Inter-comparison and calibration Workshop/experiment

September 2022

PTI TELEDetect WP6, Colmenar de Oreja (Madrid, Spain)

This workshop offered the opportunity to exchange ideas and experience in the use of spectroradiometers and cameras (field and UAV mounted sensors) for in situ and proximal sensing.

Course One Planet, One Ocean

May 2022

GEOMAR Helmholtz Centre for Ocean Research (Kiel, Germany), Online

Certificate of completion in a course on the physics, biology, and chemistry of the ocean; the impact of human interactions with the ocean on its resilience; regulations governing ocean use and misuse, along with the economics of ocean ecosystems; and sustainable solutions for coasts, fisheries, and other systems.

Course Remote Sensing as a Global Tool**July 2021***Universidad Internacional Menéndez Pelayo (UIMP), Online*

Certificate of completion in a 30-hour course providing an overview and current insights into the techniques and applications of remote sensing within the context of global change.

Course in Basic GIS and Remote Sensing**March 2021***University of Cádiz (Cádiz, Spain)*

A 40-hour fundamental training course in GIS and Remote Sensing applied to project management in marine and terrestrial environments, as well as environmental protection.

Course Hyperspectral Data for Land and Coastal Systems**February 2021***National Aeronautics and Space Administration (NASA), Online*

Certificate of completion in this training course including three sessions about the use of hyperspectral sensors on land and coastal systems.

Drone/UAV Remote Sensing**Hexacopter Drone CONDOR Certification, DroneTools****May 2023***DroneTools S.L., Gelves (Seville, Spain)*

Certificate of practical and theoretical training for authorization to operate the Condor drone by DroneTools brand.

Pix4Dmapper Essentials Workshop**June 2021***Pix4D, Online*

Received training on Pix4Dmapper Structure from Motion (SfM) photogrammetry workflows including best practices for capturing RGB images with drones and handling mapping data.

Programming**Course Introduction to Python Programming****May 2023***Spanish National Research Council (CSIC), Online*

Certificate of completion in an introductory Python programming course, covering fundamental concepts, types and data structures, flow control, functions, use of modules and packages, object-oriented programming, and error handling.

Course Python 3 from A to Z**December 2022***Frogames Support Team, Online*

A 50-hour course to learn how to use Python for real-world tasks, such as working with files and text, reading CSV or JSON files, creating data structures, loading and visualizing data needed for exploratory analysis in Data Science, and much more.

Artificial Intelligence**Intensive Technical Program in Data Science****April 2026***DATAMECUM (Spain), Online*

600-hour technical program in data science, designed to take participants from beginner level to mastery of the fundamentals of machine learning. Focused on analyzing data, predicting future outcomes, and identifying patterns using Python.

Machine learning for Earth Systems Modelling**April 2026***ECMWF, Online*

Certificate of completion for a 10-hour introductory training course on machine learning applied to weather and climate science, as well as data modeling.

Bootcamp in Data Analytics & IA**February 2026***Upgrade Hub (Spain), Online*

Certificate of completion for a 380-hour bootcamp designed to train professionals with comprehensive skills in data analysis, artificial intelligence, and interactive application development.

ML for Scientific Research: Classification and Regression Trees**August 2025***National Museum of Natural Sciences (MNCN, Spain), Online*

Certificate of completion for a 25-hour training course to learn the fundamental concepts of machine learning and the use of classification and regression trees for scientific research.

CS50's Introduction to Artificial Intelligence with Python

July 2025

Harvard University, Online

Certificate of completion for a 6-hour training course exploring the core concepts and algorithms of modern artificial intelligence, delving into the ideas that have enabled the development of video game engines, machine translation, and handwriting recognition.

AI Applied to the Environment

March 2025

DATAMECUM, University of Cádiz (Spain)

Certificate of completion for a 25-hour training course in data science and the use of artificial intelligence focused on environmental applications for scientific purposes.

The Portfolio as a Tool for Learning and Professional Development

April 2025

CSIC, Online

Certificate of completion for a 5-hour training course in addressing the integration of digital technologies in learning and the improvement of the digital competence of the researcher and the institution, through the creation of a learning portfolio that showcases the entire professional trajectory.

Google AI Essentials

March 2025

Google, Online

Certificate of completion for a 9-hour course aimed to contribute learning how to use generative AI tools to help speed up daily tasks, make more informed decisions, and develop new ideas and content.

Course MOOC Machine Learning in Weather & Climate

April 2023

European Centre for Medium-Range Weather Forecasts (ECMWF), Online

Certificate of completion on this MOOC Course exploring the basics of machine learning in weather & climate, the concepts of machine learning, and practical machine learning applications.

Emergencies

Advanced Civil Protection Course

December 2025

Spanish National Civil Protection System, Online

Certificate of completion for a 50-hour course aimed at providing participants with las competencias necesarias para facilitar el desempeño de los cometidos y responsabilidades propios de cada Órgano Directivo en lo referente a la Protección Civil.

Basic Civil Protection Course

March 2025

Spanish National Civil Protection System, Online

Certificate of completion for a 15-hour course aimed at providing participants with fundamental knowledge about the Spanish National Civil Protection System, its regulatory framework, structure, and functions, as well as an understanding of the threats we are exposed to.

Others (Hazard Prevention, Psychology, Scientific Information)

Course Psychological and Emotional Education

October 2023

ASPY Prevención (Montgat, Barcelona), Online

Certificate of completion for a 25-hour course on psychosocial factors and risks in the workplace, management and prevention of work-related stress, handling of workplace conflict situations, emotional management, and healthy leadership.

Course Field Trip Safety Resource

April 2023

Spanish National Research Council (CSIC), Online

Certificate of completion in a 50-hour basic level course in occupational hazard prevention, covering fundamental concepts of workplace safety and health, general risks and their prevention, first aid, specific risks, and prevention in field trips.

Course in Search, Management, and Communication of Scientific Information

June 2021

University of Cádiz (Cádiz, Spain)

Basic 35-hour course on searching, managing, and communicating scientific information in academic and professional contexts.

Course in Occupational Hazard Prevention in Field Work

April 2021

Spanish National Research Council (CSIC), Online

Certificate of completion in a course on Occupational Hazard Prevention in field work, focusing on procedural guidelines and the implementation of general recommendations.

Course Laboratory Practices, Waste Management, and First Aid

November 2014

University of Cádiz (Cádiz, Spain)

A 2-hour basic training course on laboratory best practices, waste management, and first aid.

Certifications

Drone training for operating in STS-01 and STS-02 scenarios

November 2025

Formadron, Granada (Spain)

Radio Operator Course

November 2023

World Aviation Flight Academy, Madrid (Spain)

Small Unmanned Aircraft System Airman Certificate (Part 107)

September 2023

Federal Aviation Administration, FAA (US)

Certificate PADI Ocean Water

December 2021

Professional Association of Diving Instructors (PADI)

Remote Drone Pilot Certificate A1/A3, A2, STS, STS01, STS02

December 2020

European Union Aviation Safety Agency, EASA

Certificate in Advanced English (CAE-C1)

February 2020

Cambridge Assessment English, Part of the University of Cambridge (UK)

Field Experience

A. Emergencies:

Las Médulas Wildfires Post-event Emergency Assessment

León (Spain), September 15th-19th 2025.

Valencia Cut-off Low Post-event Emergency Assessment

Valencia (Spain), December 8th-11th 2024.

Valencia Cut-off Low Post-event Emergency Assessment

Valencia (Spain), November 8th-11th 2024.

La Palma Volcanic Eruption

R/V Ángeles Alvariño, October 1st-3rd 2021.

B. Research Campaigns:

POLAROMICS Project Antarctic Campaign 2026

In the Spanish Antarctic Bases (BAE Gabriel de Castilla and BAE Juan Carlos I): January 31st – February 27th, 2026.

Copernicus Training In Above-Water Radiometry (FRM4SOC-2025)

On the Acqua Alta Oceanographic Tower (AAOT), July 10th-July 11th 2025 (10 hours from 8 AM to 6 PM).

ESA Advanced Ocean Synergy Training Course (OTC25) 2025

Aboard the tall ship Statsraad Lehmkuhl, April 22nd-June 3rd 2025 (41 navigation days, including 172 hours of dedicated teaching, 6 ECTS).

DICHOSO Project Antarctic Campaign 2025

Aboard Oceanographic Research vessels: BIO Hespérides, January 11th-15th 2025; B/O Sarmiento de Gamboa, February 9th-24th 2025. In the Spanish Antarctic Base: January 15th – February 9th, 2025.

Trace Metals and UV filters concentration in water sampling

Neom (Saudi Arabia), November 29th - December 8th 2024.

GEOTRACES Summer School

R/V Heincke, July 10th-15th 2022.

C. Field Operations:

Popler Island UAV/Drone Fieldwork

Popler Island (Maryland, US), October 11th 2023.

UAV/Drone Surveys

Numerous field work campaigns to conduct and support drone surveys in coastal and lacustrine environments, including more than 500 controlled UAV/drone surveys.

Other Skills

A. Computer skills:

Programing Language: Python, Matlab, R.

AI tools: ChatGPT, Gemini, Gamma, LM Studio, Claude, NotebookLM, Ollama, Streamlit, Microsoft Azure.

AI skills: RAGs, Deep-Learning, Machine Learning, LLMs.

GIS Software: SAGA GIS, QGIS, eCognition, ArcGIS.

Photogrammetric Software: Pix4D Mapper, Agisoft Metashape, Pix4Dmatic, DJI Terra.

Drone Planning Software: UgCS, Ground Station Pro, DJI Pilot.

LiDAR Analysis Software: CloudCompare, Global Mapper, LAsTools.

Image Processing: Adobe Photoshop, Adobe Lightroom, Adobe Illustrator.

B. Oceanography Sensing skills:

Biogeochemistry Sensing: Aqualogger PT, YSI EXO2, OBScope, MiniDOT LOGGER, Cyclops, RBR XR-420, NKE WiSense CTDS 300, ADCP, Pro Oceanus, Seabird HydroCAT.

Carbon Sensing: LICOR 7810.

Ocean Color Remote Sensing: TriOS RAMSES, HyperCP, ThoMaS.

C. Co-Peer Reviewing:

Scientific Journals: Journal of Hazardous Materials, Drones MPDI, Bulletin of Vulcanology, Ecological Indicators, Remote Sensing Applications: Society and Environment, Remote Sensing in Ecology and Conservation, International Journal of Remote Sensing and Remote Sensing Letters, Applied Geomatics, Geo-spatial Information Science, Frontiers in Plant Science, Remote Sensing Journal (AET "Revista de Teledetección"), Aquatic Conservation: Marine & Freshwater Ecosystems, Scientific Reports, Communications Earth & Environment, Environmental Monitoring and Assessment.

Research Projects Evaluation Pannel: ANII Uruguay (1 project proposal in 2025 call).

* Last updated: April 2026.